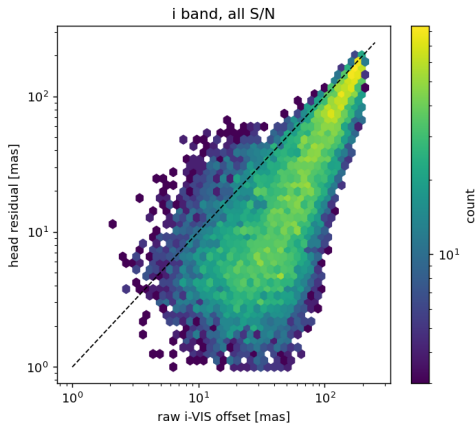
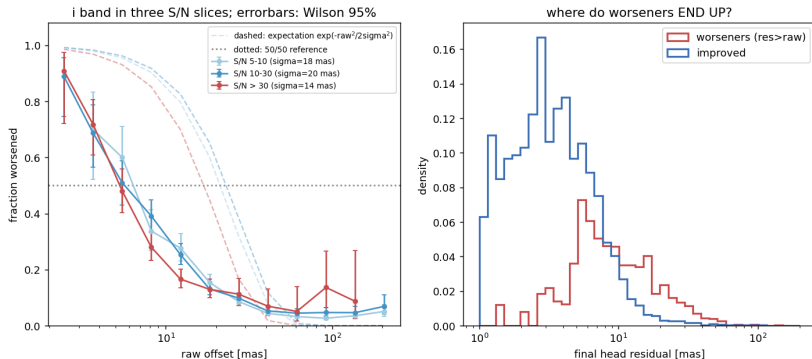


Why Astrometry Is Degraded for a Subset of Objects: Diagnosis and Cure in JAISP

July 2026 (internal, ECDFS Q1 production head)



- The head improves the population (raw ~ 48 mas $\rightarrow$ 13.4 mas median, all bands; panel: i band), but 9% of all measurements end up **farther** from VIS than they started.
- Two candidate explanations: (a) those sources were lucky before, and the small raw offset was chance; (b) the model fails on those sources specifically.



- Left: the same pattern in every S/N slice, each observed curve below its own re-measurement expectation (dashed, $P(\text{worse}) = e^{-\text{raw}^2/2\sigma^2}$); pooled over bands, shared photons preserve lucky agreement more often than chance predicts.
- Only the **bright** slice grows a tail at large raw offsets, where chance predicts zero. Right: worseners sit right of the improved because the selection forces it (res > raw truncates the floor distribution from below); the truncated-floor prediction is 21.6 mas median versus 17.8 observed. Only the excess beyond ~40 mas escapes any chance model.

For one source, per band b : fit the classical centroid, form $\text{raw}_b = \text{pos}_b - \text{pos}_{\text{VIS}}$; the head predicts a correction c_b from one shared representation; $\text{resid}_b = \text{raw}_b - c_b$. “Strongly worsened in b ” = $|\text{resid}_b| > 1.5 |\text{raw}_b|$ and worse by > 10 mas. Rate: 5.6% of bright measurements.

- **Independent between bands:** the centroid-fit noise of each band image (different exposures, different photons). If this alone drove worsening, hits on the same source would be independent coin flips: $P(3 \text{ hits}) \approx 0.056^3 \times 10 \text{ triples} \times 3185 \text{ sources} \approx 5$.
- **Shared for one source:** (1) the VIS reference position, subtracted from every band; (2) the local WCS terms (Rubin bands share Rubin’s solution, NISP its own); (3) the source itself (proper motion, colour gradients); (4) the head’s shared representation, feeding all nine corrections.

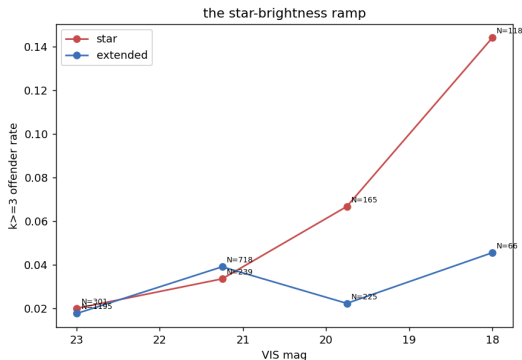
strongly worsened in $\geq k$ Rubin bands	observed	chance (permutation null)
$k \geq 2$	181	68 ± 7
$k \geq 3$	113	5 ± 2
$k \geq 4$	61	0.2
$k \geq 5$	23	0.0

The null shuffles flags within (band, raw offset, S/N) classes, preserving all rates and the VIS-induced raw correlations (hence chance $k \geq 2$ is 68, not 0). Robust to the cuts (S/N>10: 200 vs 8; all nine bands: 153 vs 12) and to the flag definition: the excess exists at every threshold. Tightening just cleans the list of chance interlopers (fiducial: 113 flagged, only ~ 6 innocent), making it clean enough to study source by source.

Stage 1 established a source-attached, band-shared cause, not yet necessarily the head. Four things are shared; each gets a measurement:

- **(1) The VIS reference?** A bad pos_{VIS} would worsen all bands at once, a plain measurement error. Gaia says no: for the offender stars the VIS labels sit **7.0 mas** from the PM-propagated Gaia position, better than the survey average.
- **(2) The WCS?** A solution error is smooth in sky position: it cannot single out one star and spare its neighbours. Offenders show **no sky clustering** (nearest-offender distance $62''$ vs $66'' \pm 5''$ for random draws), and the failure **persists in NISP** (59% of offenders worsened in all NISP bands), whose astrometric solution is independent of Rubin's.
- **(3) The source itself?** Proper motion correlates with nothing ($\rho = 0.01$ against residuals); high-PM stars are actually the head's success cases (it bridges the epochs).
- **(4) The head's representation** carries positive fingerprints, not just survival by elimination: corrections **larger than the disagreement they correct**, directions coherent across all nine bands and **both instruments** (one wrong belief applied everywhere), and Gaia places the head far from truth where every classical measurement of the same stars is at the floor (numbers on the next slides).

The two-stage verdict: repetition proves the cause is source-attached and shared; elimination plus fingerprints prove it is the head. The 113 are 0.3% of the catalog, identifiable one by one.

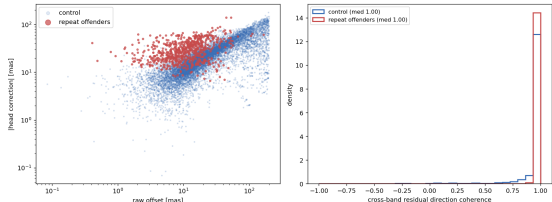


Offender rate for stars ramps 2% $\rightarrow$ **14%** from VIS 24 to VIS 17; extended sources stay flat.

Population: the 3,185 sources with ≥ 2 bright ($S/N > 20$) Rubin bands, the only ones the k statistic can score, out of 37,796 in the catalog; N labels are the per-bin denominators of that eligible population, and rate $\times$ N summed over bins and classes recovers the 113.

Further mechanisms tested and **ruled out** (beyond the WCS and proper motion of the previous slide):

- tile-edge context (paired same-source test: $+0.1$ mas)
- companions within $2''$ (13% vs 20% control)
- VIS saturation (onset ≈ 17 , masked not clipped; offender cores pristine)



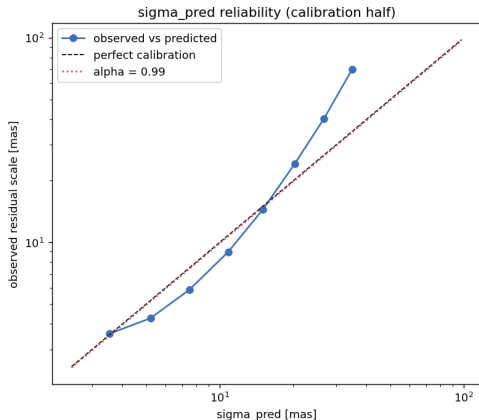
Against PM-propagated Gaia DR3 (epoch 2024.83), per star:

	VIS label	head
offender stars ($k \geq 3$)	7.0 mas	22.9 mas
all Gaia stars	9.5 mas	12.2 mas

Left: offenders receive corrections **larger than the raw disagreement** ($|c| \sim 28$ mas vs raw ~ 11 mas), coherently in every band: the head asserts a center the classical centroids contradict.

Reading: a brightness-dependent divergence between the Gaussian-fit convention and the head's feature-encoded center (PSF wings, training rarity, $\sigma_{\text{label}} = 5$ mas loss floor).

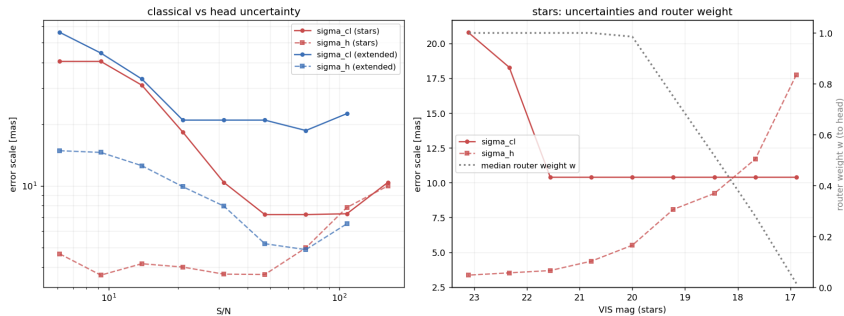
Two estimators of the same VIS-convention position: classical (error scale σ_{cl} from S/N and class) and head-corrected (the head's own σ_{pred}).



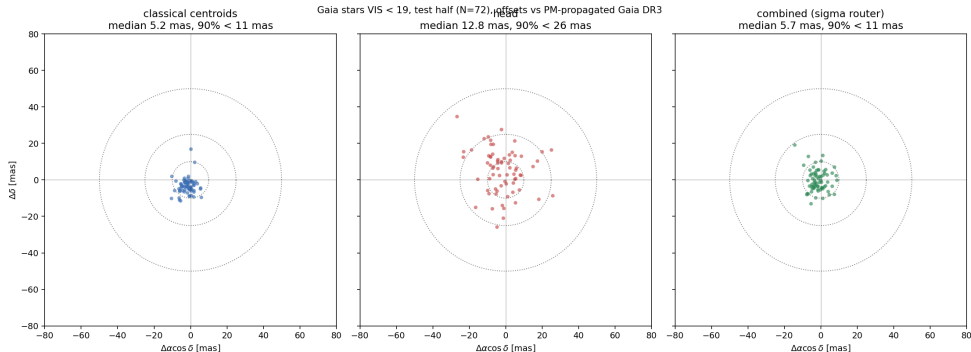
$$\text{combined} = (1 - w) \text{raw} + w \text{resid},$$

$$w = \frac{\sigma_{cl}^2 - \rho \sigma_{cl} \sigma_h}{\sigma_{cl}^2 + \sigma_h^2 - 2\rho \sigma_{cl} \sigma_h}$$

- σ_{pred} is nearly perfectly calibrated in bulk ($\alpha = 0.99$); the over-confidence lives only in the offender tail.
- The correlation matters: head and classical errors share photons ($\rho \approx 0.8$ extended, 0.5 stars); naive inverse variance costs 2 mas at the faint end, the correlation-aware weight costs nothing.
- All calibrations fit on half the tiles; all results from the other half.

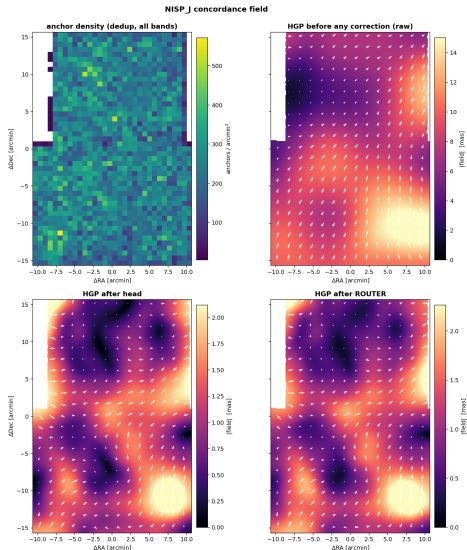


- Extended sources: classical centroiding is morphology limited ($\sigma_{cl} \gtrsim 20$ mas at every S/N); the curves **never cross**, so galaxies are pure head.
- Stars: crossing at S/N ≈ 108 (VIS ≈ 18.2); the weight (gray) slides $1 \rightarrow 0$ across VIS $20 \rightarrow 17$. The magnitude gate is **derived from the uncertainties**, not imposed.
- σ_h **rises** for bright stars: the uncertainty channel partially knows the head struggles there; the offenders are the tail where that self-awareness fails.



median radial offset [mas]	classical	head	router
all sources	50.6	14.6	14.7
bright stars, vs Gaia truth	5.2	12.8	5.4
faint (S/N < 10)	64.9	18.5	18.5

Bright stars made worse than classical: 38% (head) → 16% (router), with the large-amplitude halo gone



- Refit the field (HGP super-tight, PINN) on router residuals: median change **0.16 mas**, an order of magnitude below the field amplitude. The robust fit had already absorbed the 113 outliers; Figure 10 and the FITS products stand.
- The ~ 15 mas lobe at lower right is **real**: coherent in 8 of 9 bands relative to VIS, normal anchor coverage, both solvers agree, and it sits within MER tile TILE102043549. Eight rulers agree and one does not: a localized feature of the **VIS solution**.
- By construction no VIS-trained correction can remove it; the released raw-field FITS is precisely the map that carries it.

- Above the 1:1 line sit two populations: mostly benign regression to the corrected floor (about three quarters of worsened sources), plus real failures concentrated in bright point sources, 0.3% of the catalog under the fiducial flag.
- Gaia certifies the mechanism: the head, not the labels; a brightness-dependent centroid-convention divergence on healthy stellar profiles, with PM, edges, WCS, companions, and saturation all measured and eliminated.
- The calibrated- σ router is never worse than either estimator: head performance at the faint end, classical parity on bright stars, zero cost.
- The head improves astrometry where classical methods are noise limited and defers where they are systematics limited. That is the paper sentence.